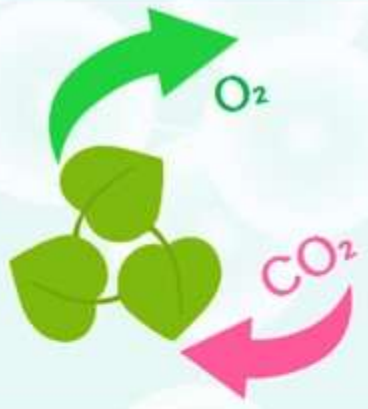


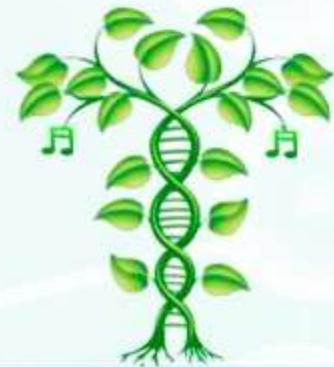


BIOLOGY



REPRODUCTION IN PLANTS -1

MODULE - 10



DO THE PLANTS REPRODUCE THE SAME WAY?



MANGO TREE



GRASSES



APPLE TREE

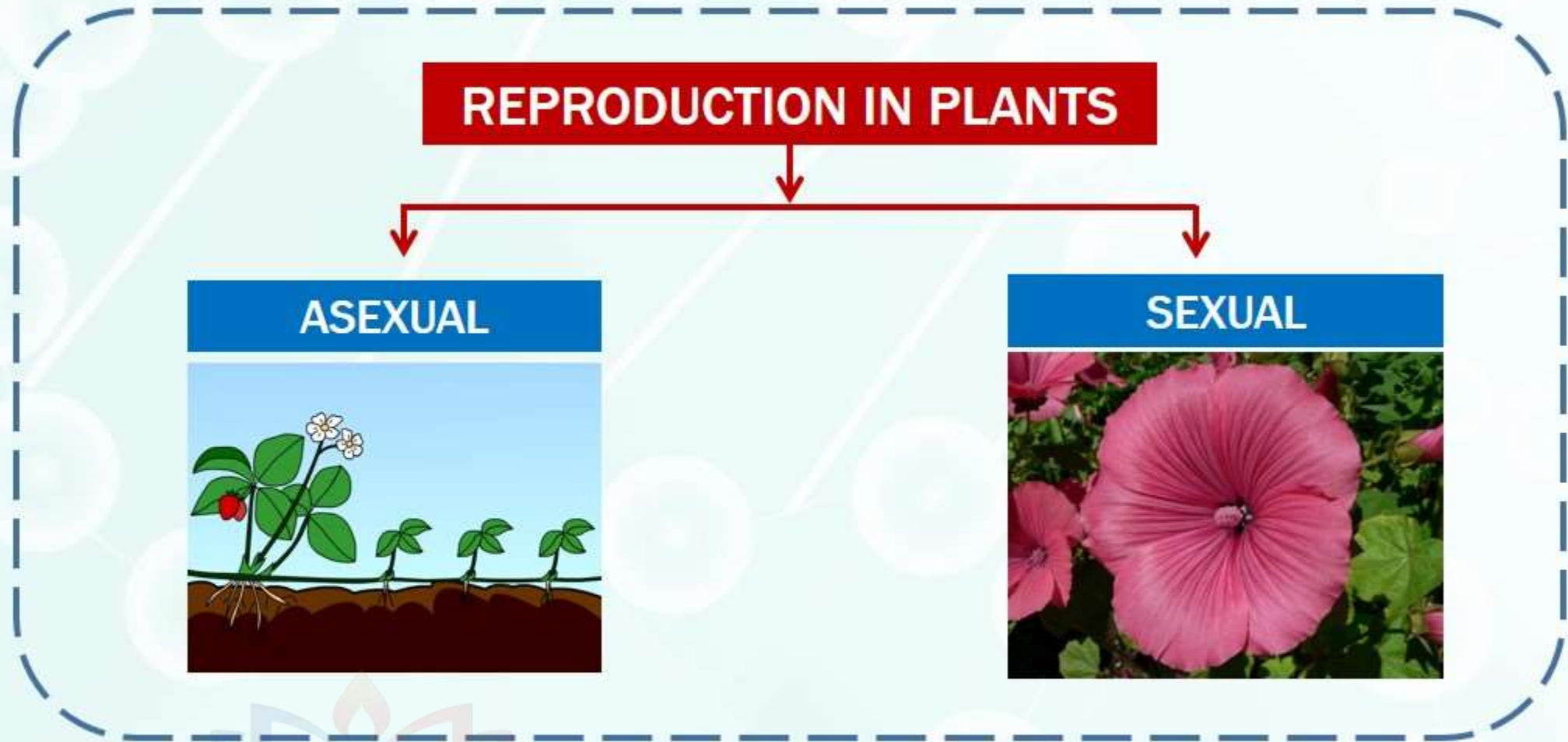


JASMINE PLANT



ROSE PLANT





*An angiosperm (flowering plant) has two stages in its life cycle:
vegetative stage and reproductive stage*



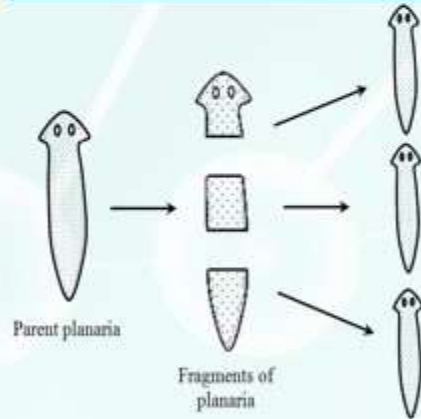
ASEXUAL REPRODUCTION

BUDDING



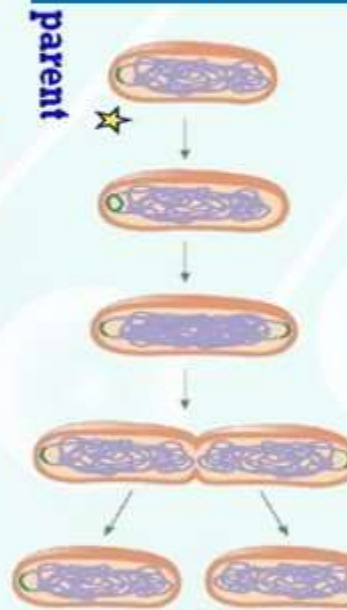
- Hydra
- Bacteria
- Yeast

FRAGMENTATION



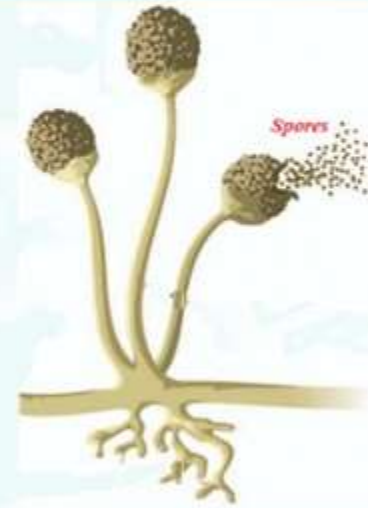
- Planaria
- Algae

FISSION



- Amoeba
- Archea
- Bacteria

SPORE FORMATION



- Fungi
- Algae
- Plants

VEGETATIVE PROPAGATION

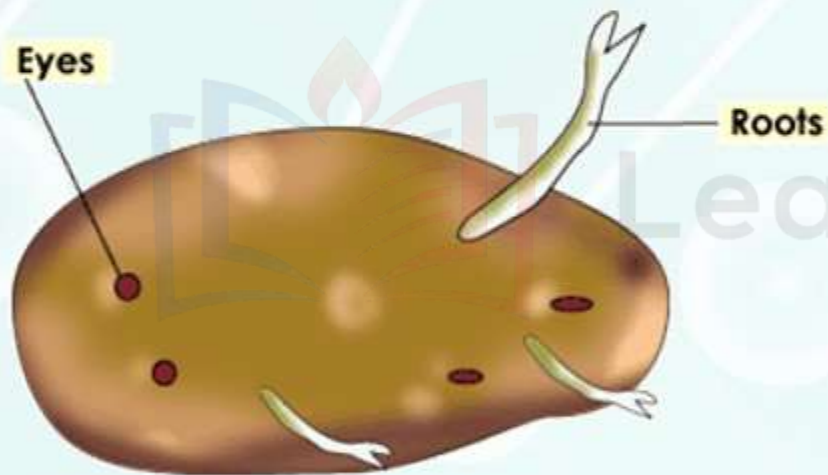


Plants





VEGETATIVE PROPAGATION IN PLANTS



- It is a form of asexual reproduction in plants. It **does not involve flowers and seeds**.
- Instead, a new plant grows from a vegetative part, like **root, stem or leaves** of the parent plant.
- Since **no gametes are involved**, the plants produced asexually have **identical genomes** and the offspring form what is known as a **clone**.
- In some cases of vegetative reproduction, the structures involved also **become storage organs** and swell with stored food, e.g. potatoes.

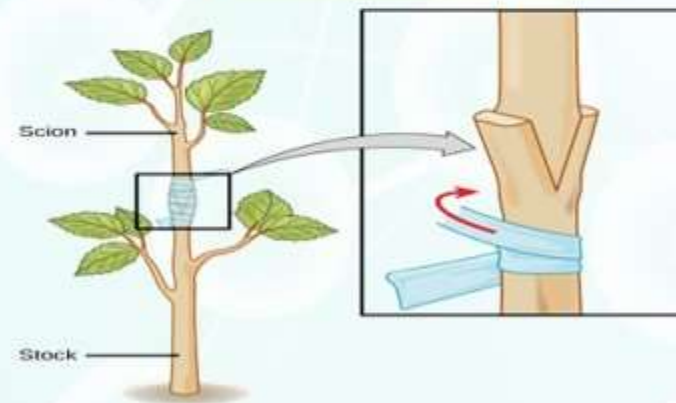


VEGETATIVE PROPAGATION IN PLANTS

NATURAL



ARTIFICIAL



VEGETATIVE PROPAGATION THROUGH UNDERGROUND ROOTS

ROOT TUBERS



Dahlia



Sweet potato



Asparagus



Cassava

Cassava or Tapioca is used to make 'sabudana'



VEGETATIVE PROPAGATION THROUGH UNDERGROUND STEMS

BULBS



- Onions
- Narcissus
- Lilly

RHIZOMES



- Ginger
- Turmeric
- Banana

TUBERS



- Potato

CORMS



- Colocasia (arvi)
- Gladiolus
- Zaminkand



VEGETATIVE PROPAGATION THROUGH SUB-AERIAL STEMS

RUNNERS



LAWN GRASS

STOLONS



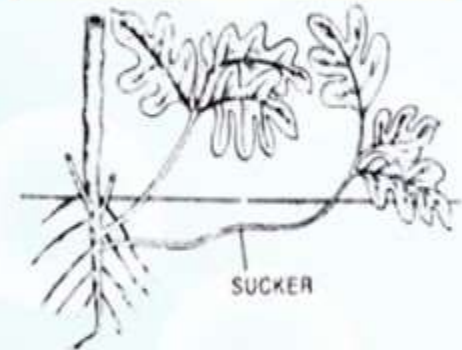
STRAWBERRY

OFFSETS



WATER HYACINTH

SUCKERS



MINT

These stems usually give rise to adventitious roots



VEGETATIVE PROPAGATION THROUGH BUDS - ADVENTITIOUS BUDS



BRYOPHYLLUM



KALANCHOE

- The leaves of a number of plants produce adventitious buds in their margin.
- These buds grow into new plants either when the leaf falls from the parent plant or sometimes even while it is still attached.



VEGETATIVE PROPAGATION THROUGH MODIFIED FLOWERS - BULBILS



AGAVE



OXALIS



GLOBBA



DIOSCOREA

*The flowers become modified into small tufts called bulbils.
These fall on the ground and grow into new plants.*



ARTIFICIAL VEGETATIVE PROPAGATION

GRAFTING



Mango

BUD GRAFTING



- Peaches
- Pears
- Plum
- Citrus

CUTTING



- Sugarcane
- Bougainvillea
- Dahlia

LAYERING



- Jasmine
- Apples
- Gooseberry
- Magnolia



ARTIFICIAL VEGETATIVE PROPAGATION

GRAFTING

The cutting of a plant (scion) is attached to the stem of another rooted plant (stock). After some time the attached cutting becomes an integrated part of the rooted plant.



BUD GRAFTING

A bud is taken along with portion of bark. The bud is fixed tightly on the stock with a tape. It gets attached to the stock after some time and new branches are formed.



CUTTING

Stem cuttings with some nodes and internodes are placed in moist soil which gives rise to adventitious roots and a new plant subsequently.



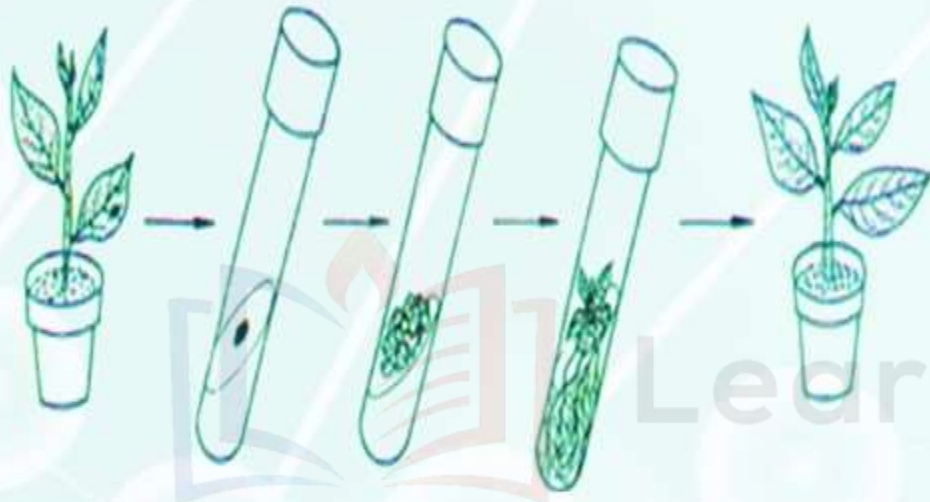
LAYERING

One or more branches are bent close to the ground and covered with moist soil. After some time, the underground portion of those branches produce new roots and develop into a separate plant.



ARTIFICIAL VEGETATIVE PROPAGATION

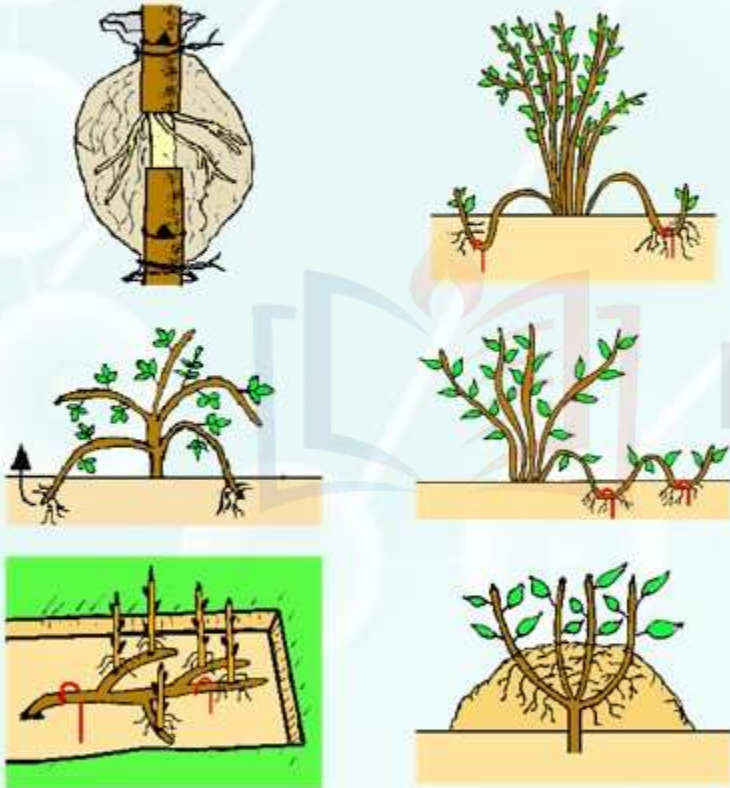
TISSUE CULTURE



- This method allows us to grow whole plant from cells taken from various parts of the plant body.
- A small piece of tissue of a desired plant is cut.
- This is placed with a **suitable nutrient medium under proper conditions**.
- The tissue grows into an unorganized mass, known as **callus**.
- Small part of this tissue is put in another medium, which induces the formation of plantlets.
- The plantlets can be transplanted in soil or pots for developing to maturity.
- This technique is also called **micro propagation**.
- Used in growing Asparagus, orchids, Chrysanthemum.



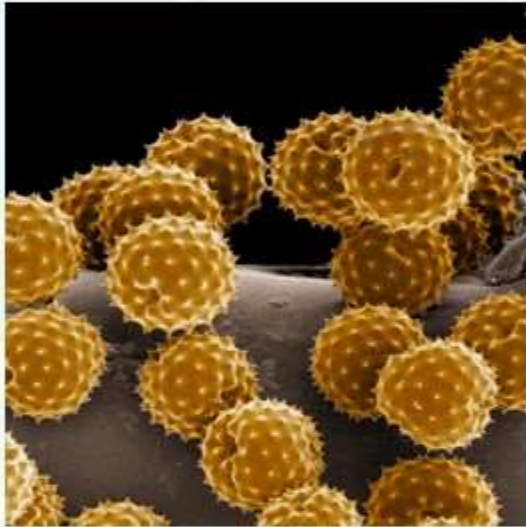
VEGETATIVE PROPAGATION - ADVANTAGES



- Plant which **do not produce seeds**, are propagated by the method e.g. Sugar-cane, banana, potato, rose, orange, seedless grapes, jasmines etc.
- Vegetative propagation is a **cheaper, easier and more rapid method of propagation in plants** than growing plants from their seeds.
- The **trait (character) of a plant is preserved** by vegetative reproduction.
- The plants generated requires less time to grow and becomes genetically similar to the parent stock.
- The plants **skip the risky seedling stage** in which a number of plants die under adverse conditions.
- However, plants grown by this method lose vigour and become prone to diseases.



SEXUAL REPRODUCTION IN PLANTS



MALE GAMETE



FEMALE GAMETE



SEED



Learning Space

WHORLS OF A FLOWERING PLANT

CALYX



COROLLA



STAMEN



PISTIL



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